

Credit Card Fraud Detection System

Practice Questions and Answers

1. What is the main objective of this project?

The main objective of this project is to develop a machine learning-based system that can detect fraudulent credit card transactions by analyzing transaction patterns and identifying unusual activities.

2. What problem does this system solve?

The system addresses the growing issue of credit card fraud by automatically identifying suspicious transactions and helping users or administrators take action before financial damage occurs.

3. Why is fraud detection important in digital payment systems?

Fraud detection is important because digital transactions are increasing rapidly. Without intelligent monitoring systems, unauthorized transactions can cause financial losses to both customers and financial institutions.

4. What type of application is this project?

This project is a **full-stack web application** that includes a frontend interface, backend server, database, and a machine learning model for fraud detection.

5. Which technologies were used in this project?

Frontend technologies include React, Tailwind CSS, and Vite.

Backend technologies include FastAPI, Python, SQLAlchemy, and SQLite.

Machine learning was implemented using scikit-learn.

6. What machine learning algorithm is used in this project?

The project uses the **Random Forest Classifier**, which is an ensemble learning algorithm that builds multiple decision trees and combines their results to improve prediction accuracy.

7. Why was Random Forest chosen for this project?

Random Forest was chosen because it provides high accuracy, reduces overfitting, works well with structured data, and handles large datasets effectively.

8. What are the main features of the system?

The main features include user registration, login, transaction simulation, fraud detection using machine learning, alerts for suspicious transactions, and an admin dashboard for monitoring system activity.

9. What roles exist in this system?

There are two roles:

User (Cardholder) and Admin.

Users can simulate transactions and monitor their activity, while administrators can monitor all users and system analytics.

10. What does the user dashboard display?

The user dashboard displays transaction history, risk classification of each transaction, and alerts for suspicious or fraudulent transactions.

11. What is the transaction simulator?

The transaction simulator allows users to simulate credit card transactions by entering transaction details such as amount, location, device type, and payment mode.

12. What happens when a transaction is simulated?

When a transaction is simulated, the system calculates features such as deviation score and sends the data to the machine learning model to predict whether the transaction is fraudulent.

13. What is a deviation score?

Deviation score measures how unusual a transaction is compared to the user's average spending.

Deviation Score = Transaction Amount / Average Spending

14. How are transactions classified in the system?

Transactions are classified into three categories:

Safe – normal transaction

Suspicious – unusual behavior detected

Fraud – high probability of fraud according to the ML model

15. What features are used by the machine learning model?

The model uses the following features:

Transaction amount

Deviation score

Transaction location

Device type

Payment mode
Transaction time

16. How does the system store data?

The system stores data in a SQLite database, which contains tables for users, transactions, and machine learning model information.

17. What is the purpose of the Cardholders table?

The Cardholders table stores user information such as name, email, mobile number, card number, and financial parameters like monthly income and daily spending limit.

18. What is the purpose of the Transactions table?

The Transactions table stores details of every simulated transaction including amount, location, device type, risk label, and timestamp.

19. What is the ML_Models table used for?

This table stores metadata about machine learning models such as model version, accuracy score, and last training date.

20. What is FastAPI?

FastAPI is a modern Python web framework used to build APIs quickly and efficiently with automatic documentation and high performance.

21. Why was FastAPI used instead of other frameworks?

FastAPI was used because it is lightweight, fast, supports asynchronous programming, and automatically generates API documentation using Swagger.

22. What is the role of React in this project?

React is used to build the frontend interface where users interact with the system through forms, dashboards, and visualizations.

23. How does the frontend communicate with the backend?

The frontend communicates with the backend using REST APIs through HTTP requests using Axios.

24. What is REST API?

REST API is a communication interface that allows the frontend to send requests and receive responses from the backend server.

25. What is the purpose of the admin dashboard?

The admin dashboard allows administrators to view all users, monitor transactions, analyze fraud statistics, and retrain the machine learning model.

26. What is the high-risk transaction trigger?

This feature allows the admin to manually create a high-risk transaction to demonstrate how the system detects fraud.

27. What happens when a fraud transaction is detected?

When fraud is detected, the transaction is labeled as Fraud and shown in the user's alerts section.

28. What is the alerts module?

The alerts module displays all suspicious and fraudulent transactions for the user so they can review and report them.

29. What is the feedback mechanism in the system?

Users can report wrong predictions if the system incorrectly classifies a transaction, which can help improve the model during retraining.

30. How is the machine learning model trained?

The model is trained using synthetic transaction data that simulates normal and fraudulent transactions.

31. Why is synthetic data used?

Synthetic data is used because real credit card transaction data is sensitive and not publicly available.

32. What evaluation metrics are used to measure model performance?

The system evaluates model performance using accuracy, precision, recall, and F1-score.

33. What is precision in fraud detection?

Precision measures how many transactions predicted as fraud are actually fraudulent.

34. What is recall in fraud detection?

Recall measures how many actual fraudulent transactions were correctly identified by the model.

35. What is F1-score?

F1-score is the harmonic mean of precision and recall, providing a balanced evaluation metric.

36. How does the system architecture work?

The system follows a client-server architecture where the frontend communicates with the backend through REST APIs, and the backend interacts with the database and machine learning model.

37. What type of database is used in this project?

The project uses SQLite, which is a lightweight file-based relational database.

38. What security features are included in this system?

Security features include password hashing using bcrypt, input validation using Pydantic, and protection against SQL injection using SQLAlchemy ORM.

39. What are the limitations of this project?

The project uses simulated transactions and synthetic training data, and it does not integrate with real banking systems.

40. What improvements can be made in the future?

Future improvements may include integration with real banking APIs, real-time transaction monitoring, deep learning models, and mobile application support.

– All the best –